# Lecture 5 (Part 2): Advanced Linear Programming

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Materials are taken from different sources with modifications. I do not claim any authorship of these materials.

# Content

- Advanced linear programming
  - Simplex method fundamentals
  - Revised simplex method
  - Bounded-variables algorithm

# Simplex Method Fundamentals

In linear programming, the feasible solution space forms a **convex set** if the line segment joining any two *distinct* feasible points also falls in the set.

An **extreme point** of the convex set is a feasible point that cannot lie on a line segment joining any two *distinct* feasible points in the set.

Actually, extreme points are the same as corner points, as used previously.

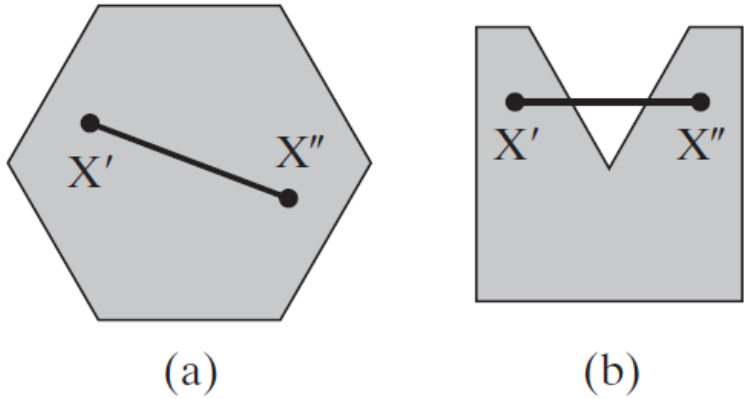



Figure illustrates two sets. Set (a) is convex (with six extreme points), and set (b) is not.

# Simplex Method Fundamentals

The graphical LP solution demonstrates that the optimum solution is always associated with a feasible extreme (corner) point of the solution space.

This result makes sense intuitively, because every feasible point in the LP solution space can be determined as a function of its feasible extreme points.

For example, in convex set (a) of Figure above, a convex combination of the extreme points, $\mathbf{X}_1, \mathbf{X}_2, \mathbf{X}_3, \mathbf{X}_4, \mathbf{X}_5$, and $\mathbf{X}_6$, identifies any feasible point $\mathbf{X}$ as

$$\mathbf{X} = \alpha_1\mathbf{X}_1 + \alpha_2\mathbf{X}_2 + \alpha_3\mathbf{X}_3 + \alpha_4\mathbf{X}_4 + \alpha_5\mathbf{X}_5 + \alpha_6\mathbf{X}_6$$
$$\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4 + \alpha_5 + \alpha_6 = 1$$
$$\alpha_i \geq 0, i = 1,2, \dots, 6$$

This observation shows that a finite number of extreme points completely define the infinite number of points in the solution space.

This result is the crux of the simplex method.

# From Extreme Points to Basic Solutions

It is convenient to express the general LP problem in equation form using matrix notation.

Define

- **X** as an $n$-vector representing the variables,
- $A$ as an $(m \times n)$ matrix representing the constraint coefficients,
- **b** as a column vector representing the right-hand side, and
- **C** as an $n$-vector representing the objective-function coefficients.

The LP is then written as

$$\text{Maximize or minimize } z = \mathbf{CX}$$

subject to

$$A\mathbf{X} = \boldsymbol{b}$$

$$\mathbf{X} \geq \mathbf{0}$$

**NOTE:** the rightmost $m$ elements of $\mathbf{X}^T$ represent the starting basic variables. Hence, the rightmost $m$ columns of **A** always form an identity matrix **I**.

# From Extreme Points to Basic Solutions

A basic solution of $A\mathbf{X} = \mathbf{b}$ is determined by setting $n - m$ variables equal to zero, and then solving the resulting $m$ equations in the remaining $m$ unknowns, provided that the resulting solution is unique.

Given this definition, the theory of linear programming establishes the following result:

$$\text{Extreme points of } \{\mathbf{X} | A\mathbf{X} = \mathbf{b}\} \Rightarrow \text{Basic solutions of } A\mathbf{X} = \mathbf{b}$$

Thus, the basic solutions of $A\mathbf{X} = \mathbf{b}$ provide all the information needed to determine the optimum solution of the LP problem.

Furthermore, the nonnegativity restriction, $\mathbf{X} \geq \mathbf{0}$ , limits the search for the optimum to the feasible basic solutions only.

# From Extreme Points to Basic Solutions

To formalize the definition of a basic solution, the system $A\mathbf{X} = \mathbf{b}$ is written in vector form as

$$\sum_{j=1}^{n} \mathbf{P}_j x_j = \mathbf{b}$$

The vector $\mathbf{P}_j$ is the $j$th column of "$A$". A subset of $m$ vectors forms a **basis**, "$B$", if, and only if, the selected $m$ vectors are **linearly independent**. In this case, the matrix $B$ is **nonsingular**.

Defining $\mathbf{X}_B$ as an $m$-vector of the basic variables, then

$$B\mathbf{X}_B = \mathbf{b}$$

# From Extreme Points to Basic Solutions

Using the inverse $B^{-1}$, the associated basic solution is

$$\mathbf{X}_B = B^{-1}\mathbf{b}$$

If $B^{-1}\mathbf{b} \geq \mathbf{0}$, then $\mathbf{X}_B$ is feasible.

The remaining $n$ - $m$ variables are **nonbasic** at zero level.

**Example**

- Determine all the basic feasible and infeasible solutions of the following system of equations:

$$\begin{bmatrix} 1 & 3 & -1 \\ 2 & -2 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

The following table summarizes the results.

| B | $B\mathbf{X}_B = \mathbf{b}$ | Solution | Type |
|---|---|---|---|
| $(\mathbf{P}_1, \mathbf{P}_2)$ | $\begin{bmatrix} 1 & 3 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$ | $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = -\frac{1}{8}\begin{bmatrix} -2 & -3 \\ -2 & 1 \end{bmatrix}\begin{bmatrix} 4 \\ 2 \end{bmatrix} = \frac{1}{4}\begin{bmatrix} 7 \\ 3 \end{bmatrix}$ | Feasible |
| $(\mathbf{P}_1, \mathbf{P}_3)$ | Not a basis because $\mathbf{P}_1, \mathbf{P}_3$ are dependent | ---------- | ---------- |
| $(\mathbf{P}_2, \mathbf{P}_3)$ | $\begin{bmatrix} 3 & -1 \\ -2 & -2 \end{bmatrix} \begin{bmatrix} x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$ | $\begin{bmatrix} x_2 \\ x_3 \end{bmatrix} = -\frac{1}{8}\begin{bmatrix} -2 & 1 \\ 2 & 3 \end{bmatrix}\begin{bmatrix} 4 \\ 2 \end{bmatrix} = \frac{1}{4}\begin{bmatrix} 3 \\ -7 \end{bmatrix}$ | Infeasible |

# Generalized Simplex Tableau in Matrix Form

This section develops the general simplex tableau in matrix form.

Consider the LP in equation form:

$$\text{Maximize or minimize } z = \mathbf{CX}$$

subject to

$$\mathbf{AX} = \boldsymbol{b}$$
$$\mathbf{X} \geq \mathbf{0}$$

Equivalently, the problem can be written as

$$\begin{bmatrix} 1 & -\mathbf{C} \\ \mathbf{0} & A \end{bmatrix} \begin{bmatrix} z \\ \mathbf{X} \end{bmatrix} = \begin{bmatrix} 0 \\ \mathbf{b} \end{bmatrix}$$

Suppose that $B$ is a feasible basis of the system $\mathbf{AX} = \boldsymbol{b}, \mathbf{X} \geq \mathbf{0}$, and let $\mathbf{X}_B$ be the corresponding vector of basic variables and $\boldsymbol{C}_B$ its associated objective vector.

# Generalized Simplex Tableau in Matrix Form

Given all the nonbasic variables are zero, the solution is then computed as

$$\begin{bmatrix} z \\ \mathbf{X}_B \end{bmatrix} = \begin{bmatrix} 1 & -\mathbf{C}_B \\ \mathbf{0} & B \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ \mathbf{b} \end{bmatrix} = \begin{bmatrix} 1 & \mathbf{C}_B B^{-1} \\ 0 & B^{-1} \end{bmatrix} \begin{bmatrix} 0 \\ \mathbf{b} \end{bmatrix} = \begin{bmatrix} \mathbf{C}_B B^{-1}\mathbf{b} \\ B^{-1}\mathbf{b} \end{bmatrix}$$

The complete simplex tableau in matrix form can be derived from the original equations as

$$\begin{bmatrix} 1 & \mathbf{C}_B B^{-1} \\ 0 & B^{-1} \end{bmatrix} \begin{bmatrix} 1 & -\mathbf{C} \\ \mathbf{0} & A \end{bmatrix} \begin{bmatrix} z \\ \mathbf{X} \end{bmatrix} = \begin{bmatrix} 1 & \mathbf{C}_B B^{-1} \\ 0 & B^{-1} \end{bmatrix} \begin{bmatrix} 0 \\ \mathbf{b} \end{bmatrix}$$

Matrix manipulations then yield the following equations:

$$\begin{bmatrix} 1 & \mathbf{C}_B B^{-1} A - \mathbf{C} \\ 0 & B^{-1} A \end{bmatrix} \begin{bmatrix} z \\ \mathbf{X} \end{bmatrix} = \begin{bmatrix} \mathbf{C}_B B^{-1}\mathbf{b} \\ B^{-1}\mathbf{b} \end{bmatrix}$$

Given the $j$th vector $\mathbf{P}_j$ of $A$, the simplex tableau column associated with variable $x_j$ can be written as

| **Basic** | $x_j$ | **Solution** |
|---|---|---|
| $z$ | $\mathbf{C}_B B^{-1}\mathbf{P}_j - c_j$ | $\mathbf{C}_B B^{-1}\mathbf{b}$ |
| $\mathbf{X}_B$ | $B^{-1}\mathbf{P}_j$ | $B^{-1}\mathbf{b}$ |

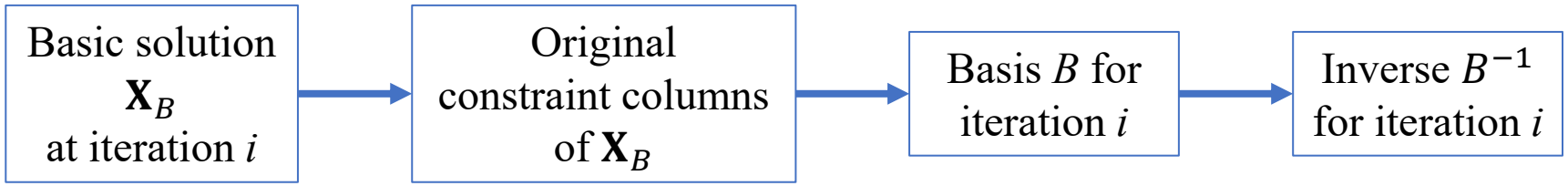
# Generalized Simplex Tableau in Matrix Form

Matrix manipulations:

$$\begin{bmatrix} 1 & \mathbf{C}_B B^{-1} - \mathbf{C} \\ 0 & B^{-1}A \end{bmatrix} \begin{bmatrix} Z \\ \mathbf{X} \end{bmatrix} = \begin{bmatrix} \mathbf{C}_B B^{-1}\mathbf{b} \\ B^{-1}\mathbf{b} \end{bmatrix}$$

**Remarks.** Look at the matrix form of the simplex tableau just given. You will notice that the inverse, $B^{-1}$, is the only element that changes from one iteration to the next— in the sense that all the other elements can be plucked directly from the original data.

That means that the *entire* tableau can be generated from the original data once the associated inverse $B^{-1}$ is known. But the matrix-format tableau reveals a deeper root for determining $B^{-1}$, namely,



# Generalized Simplex Tableau in Matrix Form

**Example**

Consider the LP in equation form:

$$\text{Maximize or minimize } z = x_1 + 4x_2 + 7x_3 + 5x_4$$

subject to

$$2x_1 + x_2 + 2x_3 + 4x_4 = 10$$
$$3x_1 - x_2 - 2x_3 + 6x_4 = 5$$
$$x_1, x_2, x_3, x_4 \geq 0$$

| Basic | $x_j$ | Solution |
|---|---|---|
| $z$ | $\mathbf{C}_B B^{-1}\mathbf{P}_j - c_j$ | $\mathbf{C}_B B^{-1}\mathbf{b}$ |
| $\mathbf{X}_B$ | $B^{-1}\mathbf{P}_j$ | $B^{-1}\mathbf{b}$ |

Generate the simplex tableau associated with the basis $B = (\mathbf{P}_1, \mathbf{P}_2)$

Given $B = (\mathbf{P}_1, \mathbf{P}_2)$, then $\mathbf{X}_B = (x_1, x_2)^T$ and $\boldsymbol{C}_B = (1,4)$. Thus

$$B^{-1} = \begin{bmatrix} 2 & 1 \\ 3 & -1 \end{bmatrix}^{-1} = \frac{1}{5}\begin{bmatrix} 1 & 1 \\ 3 & -2 \end{bmatrix}$$

We then get

$$\mathbf{X}_B = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = B^{-1}\mathbf{b} = \frac{1}{5}\begin{bmatrix} 1 & 1 \\ 3 & -2 \end{bmatrix}\begin{bmatrix} 10 \\ 5 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

To compute the constraint columns in the body of the tableau, we have

$$B^{-1}(\mathbf{P_1}, \mathbf{P_2}, \mathbf{P_3}, \mathbf{P_4}) = \frac{1}{5}\begin{bmatrix} 1 & 1 \\ 3 & -2 \end{bmatrix}\begin{bmatrix} 2 & 1 & 2 & 4 \\ 3 & -1 & -2 & 6 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 2 & 0 \end{bmatrix}$$

# Generalized Simplex Tableau in Matrix Form

**Example**

Consider the LP in equation form:

$$\text{Maximize or minimize } z = x_1 + 4x_2 + 7x_3 + 5x_4$$
$$\text{subject to}$$
$$2x_1 + x_2 + 2x_3 + 4x_4 = 10$$
$$x_1, x_2, x_3, x_4 \geq 0$$

Generate the simplex tableau associated with the basis $B = (\mathbf{P}_1, \mathbf{P}_2)$

Next, we compute the objective row as

$$\mathbf{C}_B\left(B^{-1}(\mathbf{P}_1, \mathbf{P}_2, \mathbf{P}_3, \mathbf{P}_4)\right) - C = (1,4)\begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 2 & 0 \end{bmatrix} - (1,4,7,5) = (0,0,1,-3)$$

Finally, we compute the value of the objective function as

$$z = \mathbf{C}_B B^{-1}\mathbf{b} = \mathbf{C}_B \mathbf{X}_B = (1,4)\begin{pmatrix} 3 \\ 4 \end{pmatrix} = 19$$

| Basic | $x_1$ | $x_2$ | $x_3$ | $x_4$ | Solution |
|---|---|---|---|---|---|
| $z$ | 0 | 0 | 1 | -3 | 19 |
| $x_1$ | 1 | 0 | 0 | 2 | 3 |
| $x_2$ | 0 | 1 | 2 | 0 | 4 |

| **Basic** | $x_j$ | **Solution** |
|---|---|---|
| $z$ | $\mathbf{C}_B B^{-1}\mathbf{P}_j - c_j$ | $\mathbf{C}_B B^{-1}\mathbf{b}$ |
| $\mathbf{X}_B$ | $B^{-1}\mathbf{P}_j$ | $B^{-1}\mathbf{b}$ |

# Revised Simplex Method

The iterative steps of the *revised* simplex method are *exactly the same* as in the *tableau* simplex method presented in previous.

The main difference is that the computations in the revised method are based on matrix manipulations rather than on row operations.

As such, the entire simplex tableau can be computed from the *original* data and the current inverse, thus improving the accuracy of computing $B^{-1}$ and ameliorating the machine roundoff error problem.

# Revised Simplex Algorithm

**Step 0**. Construct a starting basic feasible solution, and let B and $\boldsymbol{C_B}$ be its associated basis and objective coefficients vector, respectively.

**Step 1.** Compute the inverse $B^{-1}$ of the basis B by using an appropriate inversion method.

**Step 2.** For each nonbasic, vector $\mathbf{P}_j$ compute

$$z_j - c_j = \mathbf{C}_B B^{-1} \mathbf{P}_j - c_j$$

If $z_j - c_j \geq 0$ in maximization ($\leq 0$ in minimization) for all nonbasic vectors,

stop; the optimal solution is $\mathbf{X}_B = B^{-1}\mathbf{b}, z = \mathbf{C}_B \mathbf{X}_B$.

Else, determine the *entering* vector $\mathbf{P}_j$ having the most negative (positive) $z_j - c_j$ in case of maximization (minimization) among all nonbasic vectors.

**Step 3.** Compute $B^{-1}\mathbf{P}_j$. If all the elements of $B^{-1}\mathbf{P}_j$ are negative or zero, stop; the solution is unbounded. Else, use the ratio test to determine the *leaving* vector $\mathbf{P}_i$ .

**Step 4.** Form the next basis by replacing the *leaving* vector $\mathbf{P}_i$ with the entering vector $\mathbf{P}_j$ in the current basis $B$. Go to step 1 to start a new iteration.

**Example**

The equation form of the Tat Paint model can be expressed in matrix form as

$$\text{Maximize } z = (5, 4, 0, 0, 0, 0)(x_1, x_2, x_3, x_4, x_5, x_6)^T$$

Subject to

$$\begin{bmatrix} 6 & 4 & 1 & 0 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 \\ -1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} = \begin{bmatrix} 24 \\ 6 \\ 1 \\ 2 \end{bmatrix}$$

$$x_1, \ldots, x_6 \geq 0$$

The notation $\mathbf{C} = (c_1, c_2, \ldots, c_6)$ represents the objective-function coefficients.

$(\mathbf{P}_1, \mathbf{P}_2, \ldots, \text{and } \mathbf{P}_6)$ represent the columns vectors of the constraint equations.

The right-hand side of the constraints is the vector **b**.

**Example**

**Iteration 0**

$$\mathbf{X}_{B_0} = (x_3, x_4, x_5, x_6), \qquad \mathbf{C}_{B_0} = (0, 0, 0, 0)$$

$$B_0 = (\mathbf{P}_3, \mathbf{P}_4, \mathbf{P}_5, \mathbf{P}_6) = I, \qquad B_0^{-1} = I$$

Thus

$$\mathbf{X}_{B_0} = B_0^{-1}\mathbf{b} = (24, 6, 1, 2)^T, \qquad z = \mathbf{C}_{B_0}\mathbf{X}_{B_0} = 0$$

***Optimality computations:***

$$\mathbf{C}_{B_0}B_0^{-1} = (0, 0, 0, 0)$$

$$\{z_j - c_j\}_{j=1,2} = \mathbf{C}_{B_0}B_0^{-1}(\mathbf{P}_1, \mathbf{P}_2) - (c_1, c_2) = (-5, -4)$$

Thus, $\mathbf{P}_1$ is the entering vector.

***Feasibility computations:***

$$\mathbf{X}_{B_0} = (x_3, x_4, x_5, x_6)^T = (24, 6, 1, 2)^T$$

$$B_0^{-1}\mathbf{P}_1 = (6, 1, -1, 0)^T$$

Hence,

$$x_1 = \min\left\{\frac{24}{6}, \frac{6}{1}, -, -\right\} = \min\{4, 6, -, -\} = 4$$

and $\mathbf{P}_3$ becomes the leaving vector.

The results given above can be summarized in the familiar simplex tableau format, essentially demonstrating that the two methods are the same.

| Basic | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $x_6$ | Solution |
|---|---|---|---|---|---|---|---|
| $z$ | −5 | −4 | 0 | 0 | 0 | 0 | 0 |
| $x_3$ | 6 | | | | | | 24 |
| $x_4$ | 1 | | | | | | 6 |
| $x_5$ | −1 | | | | | | 1 |
| $x_6$ | 0 | | | | | | 2 |

**Example**

**Iteration 1**

$$\mathbf{X}_{B_1} = (x_1, x_4, x_5, x_6), \qquad \mathbf{C}_{B_1} = (5, 0, 0, 0)$$

$$B_1 = (\mathbf{P}_1, \mathbf{P}_4, \mathbf{P}_5, \mathbf{P}_6) = \begin{bmatrix} 6 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \qquad B_1^{-1} = \begin{bmatrix} 1/6 & 0 & 0 & 0 \\ -1/6 & 1 & 0 & 0 \\ 1/6 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Thus

$$\mathbf{X}_{B_1} = B_1^{-1}\mathbf{b} = (4, 2, 5, 2)^T, \qquad z = \mathbf{C}_{B_1}\mathbf{X}_{B_1} = 20$$

***Optimality computations:***

$$\mathbf{C}_{B_1}B_1^{-1} = \left(\frac{5}{6}, 0, 0, 0\right)$$

$$\{z_j - c_j\}_{j=2,3} = \mathbf{C}_{B_1}B_1^{-1}(\mathbf{P}_2, \mathbf{P}_3) - (c_2, c_3) = \left(-\frac{2}{3}, \frac{5}{6}\right)$$

Thus, $\mathbf{P}_2$ is the entering vector.

**Example**

**Iteration 1**

$$\mathbf{X}_{B_1} = (x_1, x_4, x_5, x_6), \qquad \mathbf{C}_{B_1} = (5, 0, 0, 0)$$

$$B_1 = (\mathbf{P}_1, \mathbf{P}_4, \mathbf{P}_5, \mathbf{P}_6) = \begin{bmatrix} 6 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \qquad B_1^{-1} = \begin{bmatrix} 1/6 & 0 & 0 & 0 \\ -1/6 & 1 & 0 & 0 \\ 1/6 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Thus

$$\mathbf{X}_{B_1} = B_1^{-1}\mathbf{b} = (4, 2, 5, 2)^T, \qquad z = \mathbf{C}_{B_1}\mathbf{X}_{B_1} = 20$$

***Feasibility computations:***

$$B_0^{-1}\mathbf{P}_2 = \left(\frac{2}{3}, \frac{4}{3}, \frac{5}{3}, 1\right)^T$$

Hence,

$$x_1 = \min\left\{\frac{4}{\frac{2}{3}}, \frac{2}{\frac{4}{3}}, \frac{5}{\frac{5}{3}}, \frac{2}{1}\right\} = \min\left\{6, \frac{3}{2}, 3, 2\right\} = \frac{3}{2}$$

and $\mathbf{P}_4$ becomes the leaving vector.

**Example**

**Iteration 2**

$$\mathbf{X}_{B_2} = (x_1, x_2, x_5, x_6), \qquad \mathbf{C}_{B_2} = (5, 4, 0, 0)$$

$$B_2 = (\mathbf{P}_1, \mathbf{P}_2, \mathbf{P}_5, \mathbf{P}_6) = \begin{bmatrix} 6 & 4 & 0 & 0 \\ 1 & 2 & 0 & 0 \\ -1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}, \qquad B_2^{-1} = \begin{bmatrix} 1/4 & -1/2 & 0 & 0 \\ -1/8 & 3/4 & 0 & 0 \\ 3/8 & -5/4 & 1 & 0 \\ 1/8 & -3/4 & 0 & 1 \end{bmatrix}$$

Thus

$$\mathbf{X}_{B_2} = B_2^{-1}\mathbf{b} = \left(3, \frac{3}{2}, \frac{5}{2}, \frac{1}{2}\right)^T, \qquad z = \mathbf{C}_{B_2}\mathbf{X}_{B_2} = 21$$

***Optimality computations:***

$$\mathbf{C}_{B_2}B_2^{-1} = \left(\frac{3}{4}, \frac{1}{2}, 0, 0\right)$$

$$\{z_j - c_j\}_{j=3,4} = \mathbf{C}_{B_2}B_2^{-1}(\mathbf{P}_3, \mathbf{P}_4) - (c_3, c_4) = \left(\frac{3}{4}, \frac{1}{2}\right)$$

Thus, $\mathbf{X}_{B_2}$ is optimal and the computations end.

*Summary of optimal solution:* $x_1 = 3, x_2 = 1.5, z = 21$

# Bounded-variables Algorithm

# Bounded-variables Algorithm

In LP models, variables may have explicit upper and lower bounds. For example, in production facilities, lower and upper bounds can represent the minimum and maximum demands for certain products. Bounded variables also arise prominently in solving integer programs by the branch-and-bound algorithm.

## ❑ Lower bounds

Given $\mathbf{X} \geq \mathbf{L}$, substitute $\mathbf{X} = \mathbf{L} + \mathbf{X}', \mathbf{X}' \geq 0$ throughout, and solve the problem in terms of $\mathbf{X}'$ (whose lower bound now equals zero). The original $\mathbf{X}$ is then determined by back-substitution, $\mathbf{X} = \mathbf{L} + \mathbf{X}' \geq 0$

## ❑ Upper-bounding constraints $\mathbf{X} \leq \mathbf{U}$

$$\text{Maximize } z = \{C\mathbf{X} | (A, I)\mathbf{X} = \mathbf{b}, \mathbf{0} \leq \mathbf{X} \leq \mathbf{U}\}$$

The bounded algorithm uses only the main constraints $(A, I)\mathbf{X} = \mathbf{b}, \mathbf{X} \geq \mathbf{0}$

It accounts for the upper bounds, $\mathbf{X} \leq \mathbf{U}$, implicitly by modifying the feasibility condition.

Let $\mathbf{X}_B = B^{-1}\mathbf{b}$ be a current basic feasible solution of $(A, I)\mathbf{X} = \mathbf{b}, \mathbf{X} \geq \mathbf{0}$, and assume that $\mathbf{P}_j$ is the entering vector (as determined by the regular optimality condition).

Then, given that all the nonbasic variables are zero, the constraint equation of the *i*th basic variable is

$$(\mathbf{X}_B)_i = (B^{-1}\mathbf{b})_i - \left(B^{-1}\mathbf{P}_j\right)_i x_j$$

# Bounded-variables Algorithm

$$(\mathbf{X}_B)_i = (B^{-1}\mathbf{b})_i - \left(B^{-1}\mathbf{P}_j\right)_i x_j$$

When the entering variable $x_j$ increases above zero level, $(\mathbf{X}_B)_i$ will increase or decrease depending on whether $\left(B^{-1}\mathbf{P}_j\right)_i$ is negative or positive, respectively.

Thus, in determining the value of the entering vector $\mathbf{P}_j$ , three conditions must be satisfied:

1. The basic variable remains nonnegative—that is, $(\mathbf{X}_B)_i \geq 0$ .

2. The basic variable $(\mathbf{X}_B)_i$ does not exceed its upper bound—that is, $(\mathbf{X}_B)_i \leq (\mathbf{U}_B)_i$ , where $\mathbf{U}_B$ comprises the elements of $\mathbf{U}$ corresponding to $\mathbf{X}_B$.

3. The entering variable $x_j$ cannot assume a value larger than its upper bound—that is, $x_j \leq u_j$, where $u_j$ is the *j*th element of $\mathbf{U}$.

The first condition $(\mathbf{X}_B)_i \geq 0$ is the same as in the regular simplex method. It yields

$$x_j \leq \theta_1 = \min_i \left\{ \frac{(B^{-1}b)_i}{\left(B^{-1}\mathbf{P}_j\right)_i} \middle| \left(B^{-1}\mathbf{P}_j\right)_i > 0 \right\}$$

The second condition $(\mathbf{X}_B)_i \leq (\mathbf{U}_B)_i$ specifies that

$$(B^{-1}b)_i - \left(B^{-1}\mathbf{P}_j\right)_i x_j \leq (\mathbf{U}_B)_i$$

It is satisfied if

$$x_j \leq \theta_2 = \min_i \left\{ \frac{(B^{-1}b)_i - (\mathbf{U}_B)_i}{\left(B^{-1}\mathbf{P}_j\right)_i} \middle| \left(B^{-1}\mathbf{P}_j\right)_i < 0 \right\}$$

Combining the three restrictions, $x_j$ enters the solution at the level that satisfies all three conditions—that is,

$$x_j = \min\{\theta_1, \theta_2, u_j\}$$

# Bounded-variables Algorithm

The change of basis for the next iteration depends on whether $x_j$ enters the solution at level $\theta_1, \theta_2, or\ u_j$. Assuming that $(\boldsymbol{X}_B)_r$ is the leaving variable, then we have the following rules:

1. $x_j = \theta_1 : (\boldsymbol{X}_B)_r$ leaves the basic solution (becomes nonbasic) at level zero. The new iteration is generated using the regular simplex method with *xj* and $(\boldsymbol{X}_B)_r$ as the entering and the leaving variables, respectively.

2. $x_j = \theta_2 : (\boldsymbol{X}_B)_r$ becomes nonbasic *at its upper bound.* The new iteration is generated as in the case of $x_j = \theta_1$, with one modification that accounts for the fact that $(\boldsymbol{X}_B)_r$ will be nonbasic at *upper bound.* Because the values of $\theta_1$ and $\theta_2$ require *all nonbasic variables to be at zero level* (convince yourself that this is the case!), the new nonbasic $(\boldsymbol{X}_B)_r$ at upper bound is converted to a nonbasic variable at zero level. This is achieved by using the substitution $(\boldsymbol{X}_B)_r = (\boldsymbol{U}_B)_r - (\boldsymbol{X'}_B)_r$, where $(\boldsymbol{X'}_B)_r \geq 0$. It is immaterial whether the substitution is made before or after the new basis is computed.

3. $x_j = u_j$: The basic vector $\boldsymbol{X}_B$ remains unchanged because $x_j = u_j$ stops short of forcing any of the current basic variables to reach its lower $(= 0)$ or upper bound. This means that $x_j$ will remain nonbasic *but at upper bound. The only change needed in the tableau is to use the substitution* $x_j = u_j - x_j'$ to ensure that all nonbasic variables are at zero level.

**Example**

Consider the LP in equation form:

$$\text{Maximize } z = 3x_1 + 5y + 2x_3$$

subject to

$$+y + 2x_3 \leq 14$$
$$2x_1 + 4y + 3x_3 \leq 43$$
$$0 \leq x_1 \leq 4, \qquad 7 \leq y \leq 10, \qquad 0 \leq x_3 \leq 3$$

The lower bound on $y$ is accounted for using the substitution $y = x_2 + 7$, where $0 \leq x_2 \leq 10 - 7 \leq 3$

**Iteration 0**

| Basic | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | Solution |
|---|---|---|---|---|---|---|
| $z$ | -3 | -5 | -2 | 0 | 0 | 35 |
| $x_4$ | 1 | 1 | 2 | 1 | 0 | 4 |
| $x_5$ | 2 | 4 | 3 | 0 | 1 | 15 |

**Example – cntd.**

We have $B = B^{-1} = I$, and $\mathbf{X}_B = (x_4, x_5)^T = B^{-1}\mathbf{b} = (7, 15)^T$

Given that $x_2$ is the entering variable $(z_2 - c_2 = -5)$, we get $B^{-1}\mathbf{P}_2 = (1, 4)^T$, which yields

$$\theta_1 = \min\left\{\frac{7}{1}, \frac{15}{4}\right\} = 3.75 \text{ corresponding to } x_5$$

$$\theta_2 = \infty \text{ (because all the elements of } B^{-1}\mathbf{P}_2 > 0)$$

Next, given the upper bound on the entering variable, $x_2 \leq 3$, it follows

$$x_2 = \min\{3.75, \infty, 3\} = 3$$

Because $x_2 = u_2$, $\mathbf{X}_B$ remains unchanged, and $x_2$ becomes nonbasic at its upper bound.

The substitution $x_2 = 3 - \acute{x}_2$ yields the following new tableau:

| Basic | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | Solution |
|---|---|---|---|---|---|---|
| $z$ | -3 | 5 | -2 | 0 | 0 | 50 |
| $x_4$ | 1 | -1 | 2 | 1 | 0 | 4 |
| $x_5$ | 2 | -4 | 3 | 0 | 1 | 3 |

**Example – cntd.**

**Iteration 1**

The entering variable is $x_1$. The basis vector are the same as iteration 0.
Given $B^{-1}\mathbf{P_1} = (1, 2)^T$

$$\theta_1 = \min\left\{\frac{4}{1}, \frac{3}{2}\right\} = 1.5 \text{ corresponding to } x_5$$

$$\theta_2 = \infty \text{ (because all the elements of } B^{-1}\mathbf{P}_1 > 0)$$

Thus

$$x_1 = \min\{1.5, \infty, 4\} = 1.5$$

Because $x_1 = \theta_1$, the entering variable $x_1$becomes basic, and the leaving variable $x_5$ becomes nonbasic at zero level, which yields

| Basic | $x_1$ | $x'_2$ | $x_3$ | $x_4$ | $x_5$ | Solution |
|---|---|---|---|---|---|---|
| $z$ | 0 | -1 | 5/2 | 0 | 3/2 | 109/2 |
| $x_4$ | 0 | 1 | ½ | 1 | -1/2 | 5/2 |
| $x_1$ | 1 | -2 | 3/2 | 0 | 1/2 | 3/2 |

**Example – cntd.**

**Iteration**

The new inverse is $B^{-1} = \begin{bmatrix} 1 & -1/2 \\ 0 & 1/2 \end{bmatrix}$

Next, $\mathbf{X}_B = (x_4, x_1)^T = B^{-1}\mathbf{b}' = \left(\frac{5}{2}, \frac{3}{2}\right)^T$ as computed at the end of iteration 0.

We select $x'_2$ as the entering variable, and noting that $\mathbf{P}'_2 = -\mathbf{P}_2$, we get $B^{-1}\mathbf{P}'_2 = (1, -2)^T$

Thus,

$$\theta_1 = \min\left\{\frac{5}{2}, -\right\} = 2.5 \text{ corresponding to } x_4$$

$$\theta_2 = \min\left\{-, \frac{\frac{3}{2}-4}{-2}\right\} = 1.25 \text{ corresponding to } x_1$$

We then have

$$x'_2 = \min\{2.5, 1.25, 3\} = 1.25$$

Because becomes nonbasic at upper bound resulting in the substitution .

The new tableau is

| Basic | $x'_1$ | $x'_2$ | $x_3$ | $x_4$ | $x_5$ | Solution |
|---|---|---|---|---|---|---|
| $z$ | 0 | -1 | 5/2 | 0 | 3/2 | 109/2 |
| $x_4$ | 0 | 1 | ½ | 1 | -1/2 | 5/2 |
| $x'_1$ | -1 | -2 | 3/2 | 0 | 1/2 | -5/2 |

**Example – cntd.**

Next, the entering variable $x_2'$ becomes basic and the leaving variable $x_1'$ becomes nonbasic, which yields

| Basic | $x'_1$ | $x'_2$ | $x_3$ | $x_4$ | $x_5$ | Solution |
|---|---|---|---|---|---|---|
| $z$ | ½ | 0 | 7/4 | 0 | 5/4 | **223/4** |
| $x_4$ | -1/2 | 0 | 5/4 | 1 | -1/4 | 5/4 |
| $x'_2$ | 1/2 | 1 | -3/4 | 0 | -1/4 | 5/4 |

The last tableau is feasible and optimal.

Note that the last two steps could have been reversed—meaning that we could first make $x'_2$ basic and then apply the substitution $x_1 = 4 - x_1'$ = (try it!). The sequence presented here involves less computation, however.

The optimal values of $x_1, x_2,$ and $x_3$ are obtained by back-substitution as $x_1 = u_1 - x_1' = = 4 - 0 = 4, x_2 = u_2 - x_2' = = 3 - \frac{5}{4} = \frac{7}{4}$, and $x_3 = 0$.

Finally, we get $y = l_2 + x_2 = 7 + \frac{7}{4} = \frac{35}{4}$. The associated optimal value of the objective function is $\frac{223}{4}$.